FACTORY-TO-CUSTOMER SHIPPING ROUTE EFFICIENCY ANALYSIS FOR NASSAU CANDY DISTRIBUTOR

A Data-Driven Analysis of Sales, Profitability, Shipping Lead Time and Distribution Route Efficiency

Data Science Internship Research Paper

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Technologies : Python • Pandas • NumPy • Matplotlib • Seaborn • Plotly • Streamlit

Abstract

This thesis presents an end-to-end data analytics study of the Nassau Candy Distributor dataset. The supplied data contain 10,194 records, 8,549 unique orders, 15 products, 4 customer regions and 4 shipping modes. The analysis covers data quality, sales, gross profit, products, factories, regions, states, shipping modes and factory region routes. Python-based preprocessing was followed by exploratory data analysis, feature engineering, grouped aggregation and interactive dashboard development. A derived Shipping Lead Time variable was calculated as Ship Date minus Order Date. Its minimum is 904 days, maximum 1642 days, mean 1320.84 days and median 1274 days; every record exceeds 365 days. The paper therefore treats this variable as a calculated date gap requiring source-system validation rather than as a validated operational delivery-time measure. The analysis reports total sales of $141,783.63, gross profit of $93,442.80, 38,654 units and a gross-profit-to-sales ratio of 65.91%. Route-level analysis combines shipment volume, sales, profitability and calculated lead-time measures to identify combinations that warrant further investigation. The final Streamlit dashboard provides an interactive interface for exploring the same analytical results.

Keywords

Business Analytics; Distribution Analytics; Exploratory Data Analysis; Data Quality; Route Analysis; Data Visualization; Streamlit Dashboard.

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1 Introduction

Business distribution systems generate data across orders, products, customers, locations, shipping methods and financial measures. A useful analytical workflow must therefore combine commercial and operational perspectives while first establishing whether the underlying fields are suitable for interpretation.

This project applies exploratory data analysis to the Nassau Candy Distributor dataset. The analysis moves from data-quality checks to sales and profitability summaries, then to product, factory, region, state, shipping-mode and route views. The same analytical logic is made available through an interactive Streamlit dashboard.

Exploratory Data Analysis (EDA) is particularly appropriate because it supports graphical and numerical investigation before stronger conclusions are drawn (Tukey, 1977). The project therefore emphasizes transparent summaries, distributions and grouped comparisons rather than predictive claims.

The research question is: How can descriptive data analytics and interactive visualization be used to identify commercial and distribution patterns in the Nassau Candy dataset while explicitly accounting for data-quality limitations?

2 Problem Definition

The project addresses the need to understand commercial performance and distribution patterns from a transactional dataset. Relevant questions include which products and factories contribute most to sales and profit, how activity differs across customer regions and geographic units, how shipping modes are distributed, and which factory-region combinations merit further operational review.

The objectives are to (1) clean and validate the supplied data, (2) calculate commercial and operational measures, (3) engineer a route-level analytical layer, (4) visualize important patterns, and (5) deliver an interactive dashboard that supports drill-down analysis.

The scope is descriptive and diagnostic. The study does not attempt to establish causal relationships or predict future performance. Its conclusions are limited to what the supplied dataset and project-defined analytical mappings support.

3 Literature Review

3.1 Exploratory Data Analysis

Tukey (1977) established Exploratory Data Analysis as a systematic approach for discovering patterns through graphical and numerical techniques. In a distribution context, EDA can reveal concentration, variation and unusual observations before operational metrics are interpreted. The current project follows this approach through distributions, boxplots, grouped bar charts and route-level comparisons.

Data visualization also serves a communication function. Rather than relying only on summary statistics, the notebook preserves visual evidence for sales, gross profit, shipping lead time, product, factory, regional and route patterns.

3.2 Data Quality in Analytical Systems

Data quality is multidimensional. Fox, Levitin and Redman (1994) distinguish dimensions such as accuracy, completeness and consistency, while later data-quality frameworks also emphasize validity and timeliness. This distinction is important in the present study because the dataset contains no missing values or duplicate rows, yet the date relationship produces an unusual calculated lead-time distribution.

The project therefore treats completeness as only one aspect of quality. The meaning and validity of Order Date and Ship Date must be confirmed before Shipping Lead Time is used as a conventional service KPI.

3.3 Distribution and Route Analytics

Distribution analytics connects transactions to geography, product and operational attributes. Grouping by route can reveal differences in shipment volume, sales, profitability and operational measures. In this project, a route is operationalized as a Factory × Region combination, creating a consistent unit for comparison without claiming that the route is a physical transport path.

The route layer is intended as an investigation framework. Median-based thresholds are used to identify combinations that differ from the dataset-wide distribution, but these thresholds are exploratory rather than contractual service levels.

3.4 Business Intelligence Dashboards

Business intelligence dashboards convert analytical outputs into an interactive decision-support interface. The project uses Streamlit to expose summary KPIs, filters and analytical sections, allowing users to move from overall performance to route, geographic, shipping and product/factory views.

The dashboard is not treated as a replacement for source-system validation. Its role is to make patterns visible and reproducible while displaying the date-quality limitation prominently.

4 Data and Methodology

4.1 Dataset

The dataset contains 18 original fields covering identifiers, dates, shipping mode, customer information, geography, division, region, product information, sales, units, gross profit and cost. The supplied file contains 10,194 records and 8,549 unique orders. There are 15 products, four customer regions and four shipping modes.

4.2 Data Preparation

Order Date and Ship Date were converted to datetime values using day-first parsing. Missing-value counts were calculated for all original fields, and duplicate rows were checked. The resulting dataset contains zero missing values across the original columns and zero duplicate rows.

The State/Province field is retained as supplied. Because it includes both U.S. states and Canadian provinces, geographic results are described as geographic administrative units rather than assuming that every observation is a U.S. state.

4.3 Feature Engineering

Shipping Lead Time is calculated as Ship Date minus Order Date in days. Descriptive statistics include minimum, maximum, mean, median and standard deviation. A quality flag is used for observations above 365 days. A Delayed flag is defined as Shipping Lead Time greater than the overall median of 1,274 days for exploratory route-delay analysis.

The notebook also calculates route-level measures such as shipment share, lead-time coefficient of variation, delay frequency, efficiency score and investigation flags. These derived measures are analytical constructs and not official business KPIs.

4.4 Factory Mapping

The original dataset does not include a factory field. The notebook creates a project-defined PRODUCT_FACTORY_MAP assigning the 15 products to five factory labels: Lot's O Nuts, Wicked Choccy's, Sugar Shack, Secret Factory and The Other Factory. All mapped products receive a factory value.

Because this mapping is introduced by the project rather than supplied as an original data field, factory-level findings should be validated against an authoritative business source before being used operationally.

4.5 Route Analysis

Route analysis groups observations by Factory and Region. For each combination, the notebook calculates shipments, unique orders, units, sales, cost, gross profit, average and median lead time, minimum and maximum lead time, standard deviation, delay frequency and additional derived indicators. Investigation flags combine high-volume, relatively long-lead-time and high-delay-frequency conditions.

The analysis deliberately uses relative thresholds rather than presenting an arbitrary business target. This makes the framework useful for exploratory review while avoiding an unsupported claim that a specific number of days is an acceptable or unacceptable service level.

4.6 Visualization and Dashboard Development

The executed notebook contains 20 embedded visualizations. They cover shipping lead-time distributions, sales and gross-profit distributions, product performance, factory performance, customer regions, shipping modes, geographic units and route analysis. The dashboard extends these results through interactive filters and drill-down sections.

The final paper places selected figures within the Results section and reproduces the complete notebook figure set in Appendix A.

5 Results

5.1 Dataset Quality

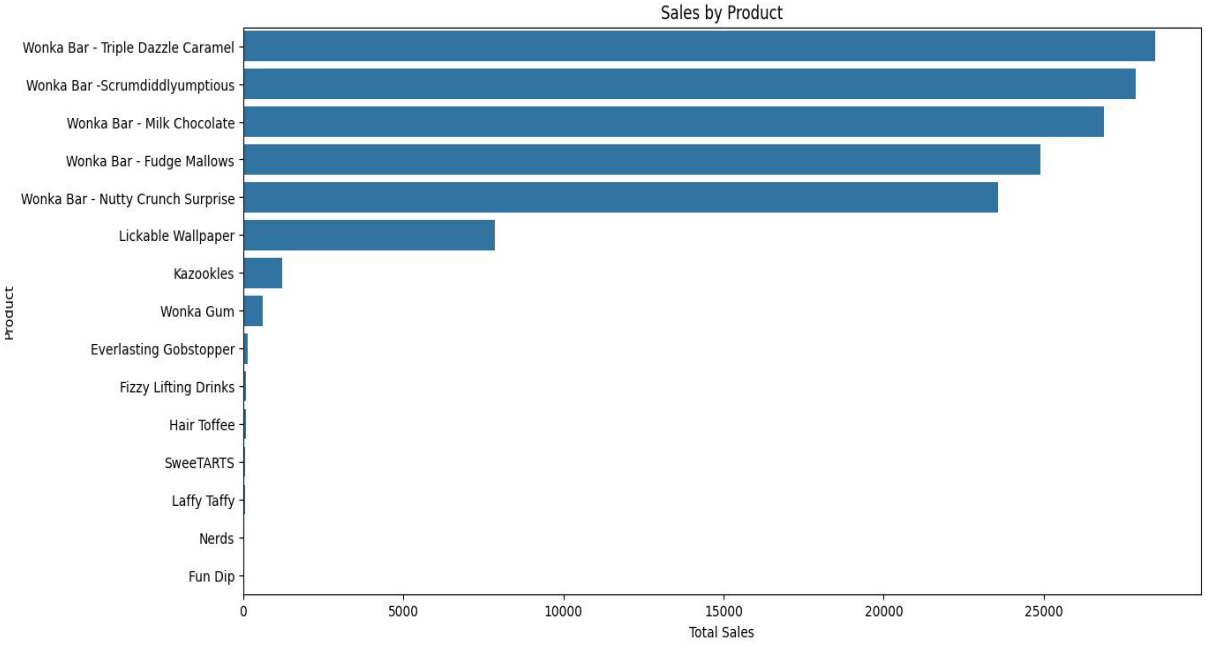
The dataset contains 10,194 records and 18 original columns. The analysis reports 0 missing values and 0 duplicate rows. This provides a complete and non-duplicated starting table, but completeness does not guarantee semantic validity of derived measures.

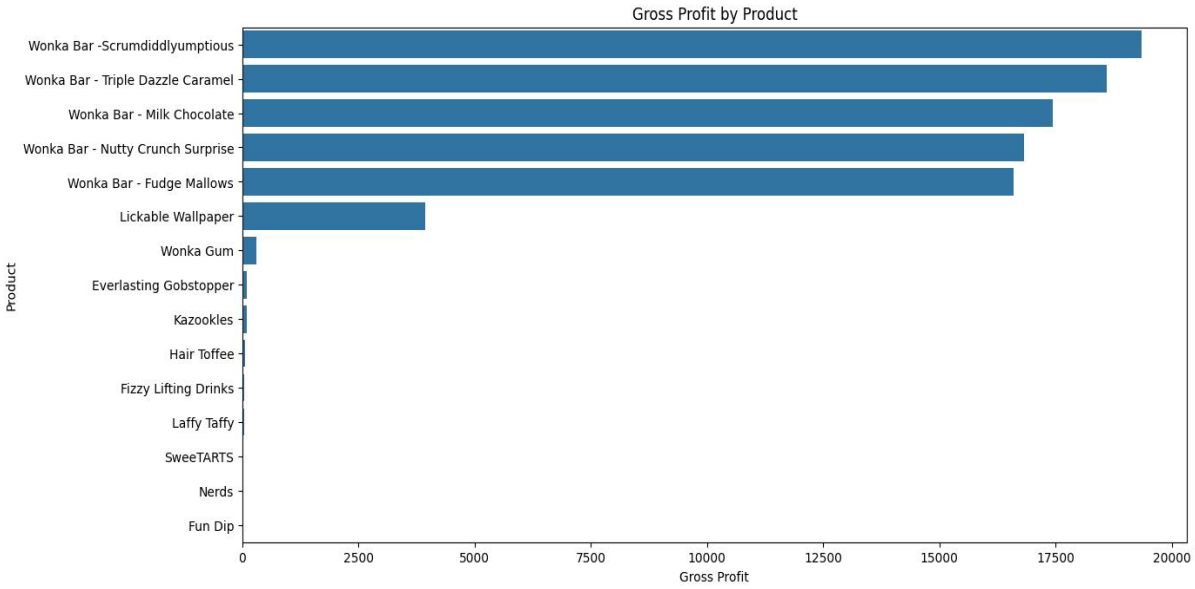
5.2 Overall Commercial Performance

Total sales are $141,783.63, gross profit is $93,442.80, units are 38,654, and unique orders are 8,549. The gross profit-to-sales ratio is 65.91%. These measures form the commercial baseline for the subsequent dimension level analysis.

5.3 Product Analysis

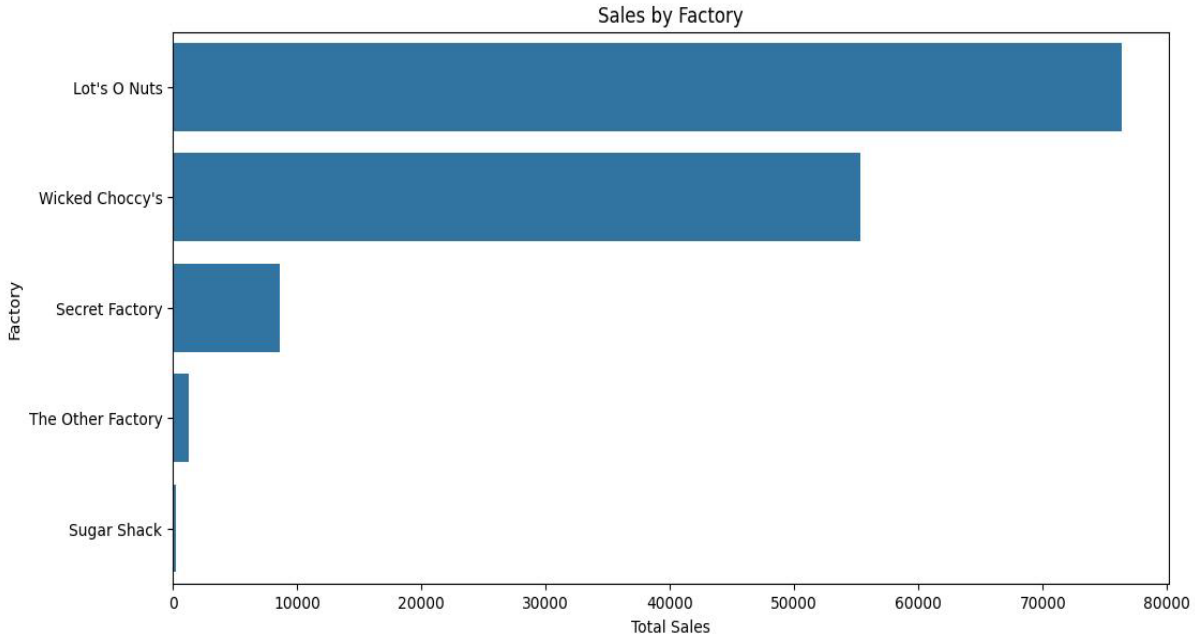
Sales are concentrated among the Wonka Bar products. The highest-sales product in the supplied data is Wonka Bar - Triple Dazzle Caramel, followed by Wonka Bar - Scrumdiddlyumptious, Wonka Bar - Milk Chocolate, Wonka Bar - Fudge Mallows and Wonka Bar - Nutty Crunch Surprise. The complete product-level visual is reproduced.





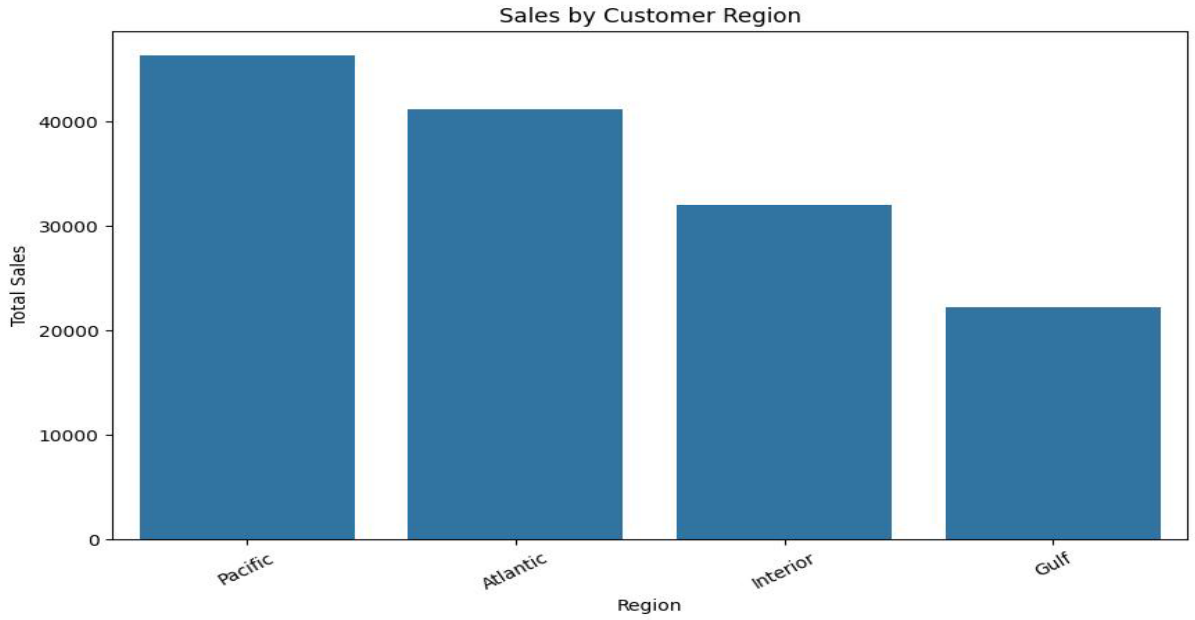
5.4 Factory Analysis

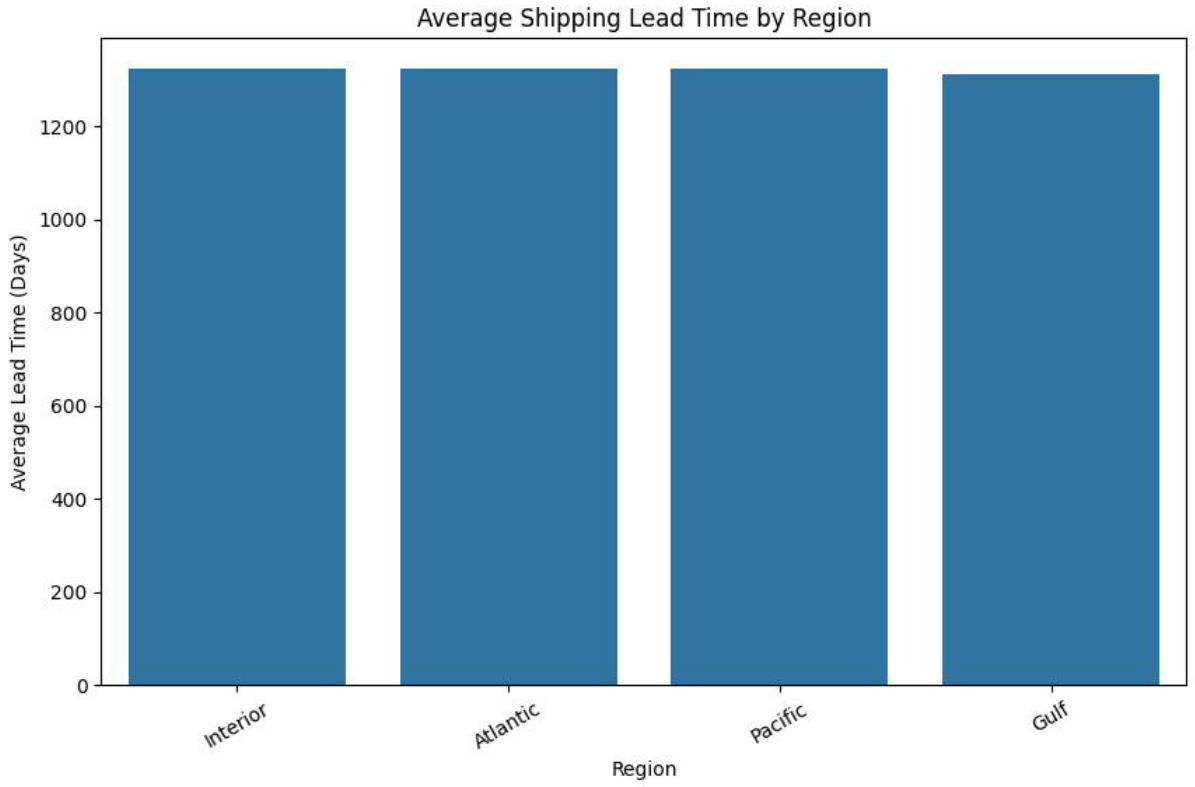
The project-defined factory mapping produces five analytical factories. Lot's O Nuts accounts for the largest shipment and sales volume, followed by Wicked Choccy's. The remaining factories represent smaller mapped product groups.

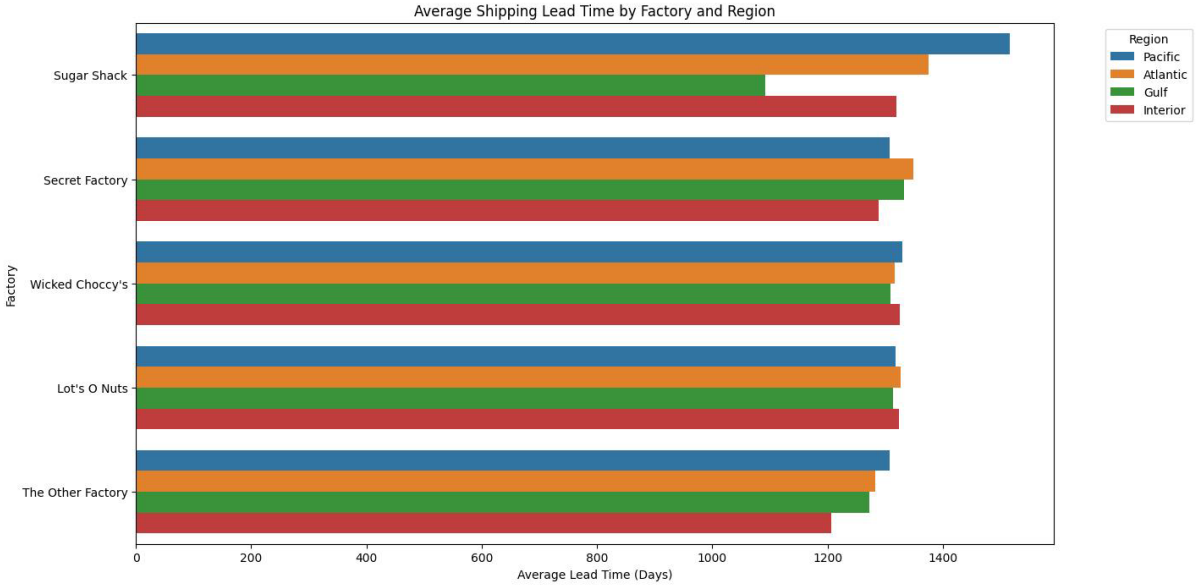


5.5 Regional Analysis

Pacific has the highest sales among the four customer regions, followed by Atlantic, Interior and Gulf. The regional differences are useful for descriptive segmentation, while the relatively similar calculated lead-time values should be interpreted cautiously because of the date-quality issue.

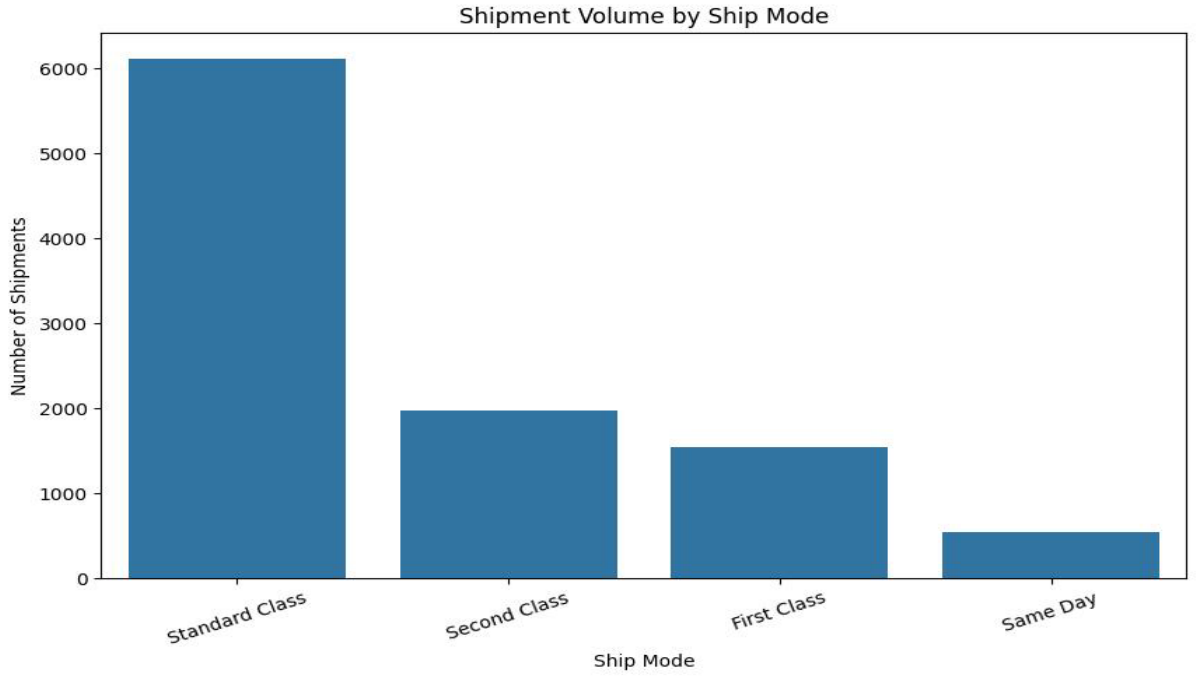


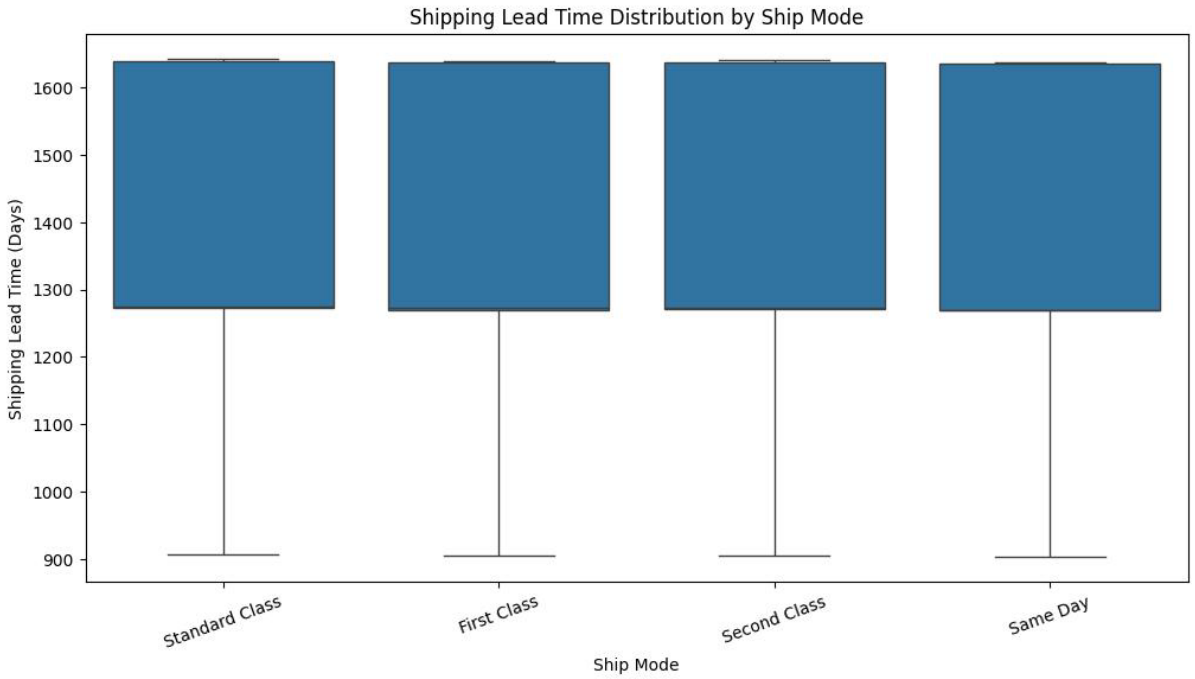


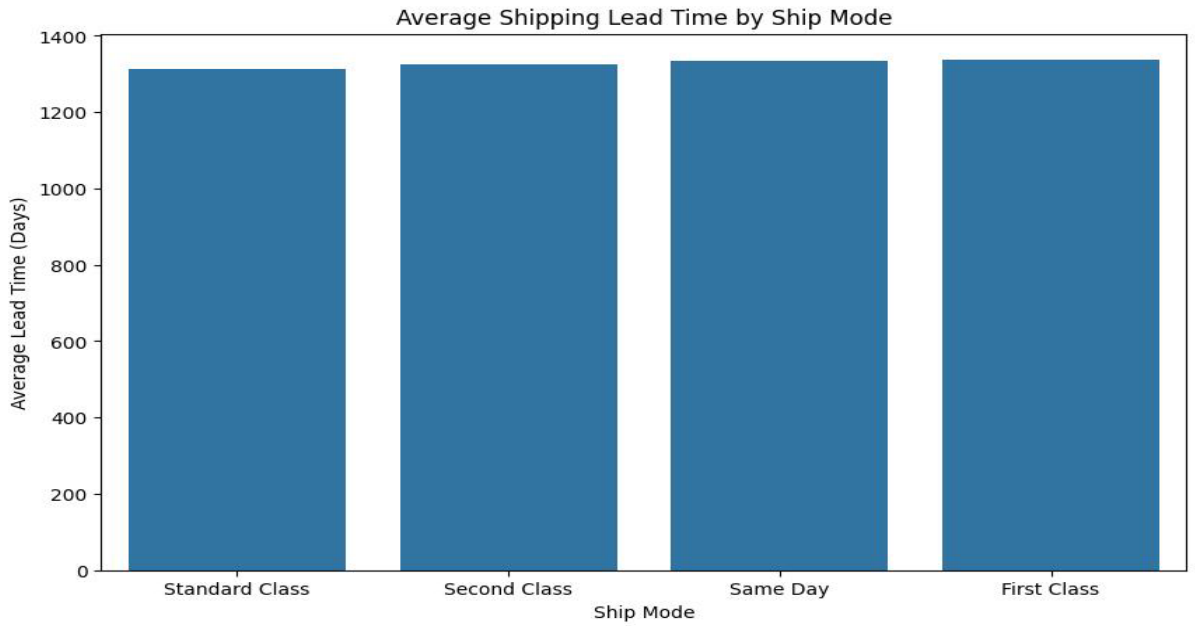


5.6 Shipping Mode Analysis

Standard Class is the most frequent shipping mode and also has the highest sales. First Class has the highest average calculated lead time among the four modes. These results describe the supplied records and should not be interpreted as evidence that one shipping mode causes longer delivery performance.

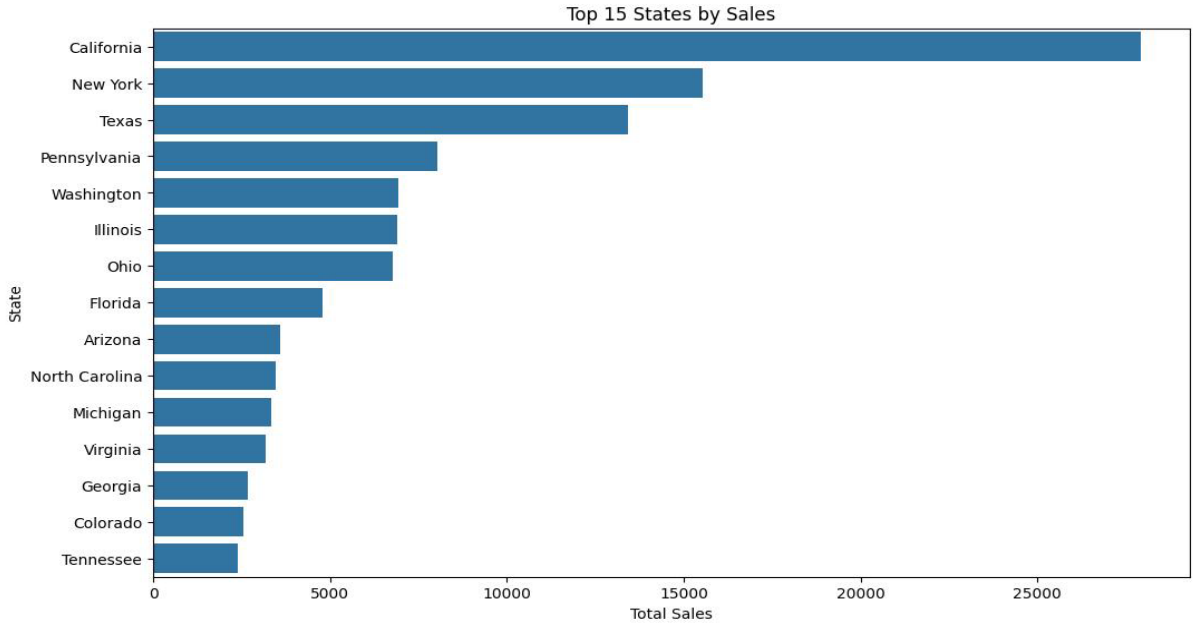


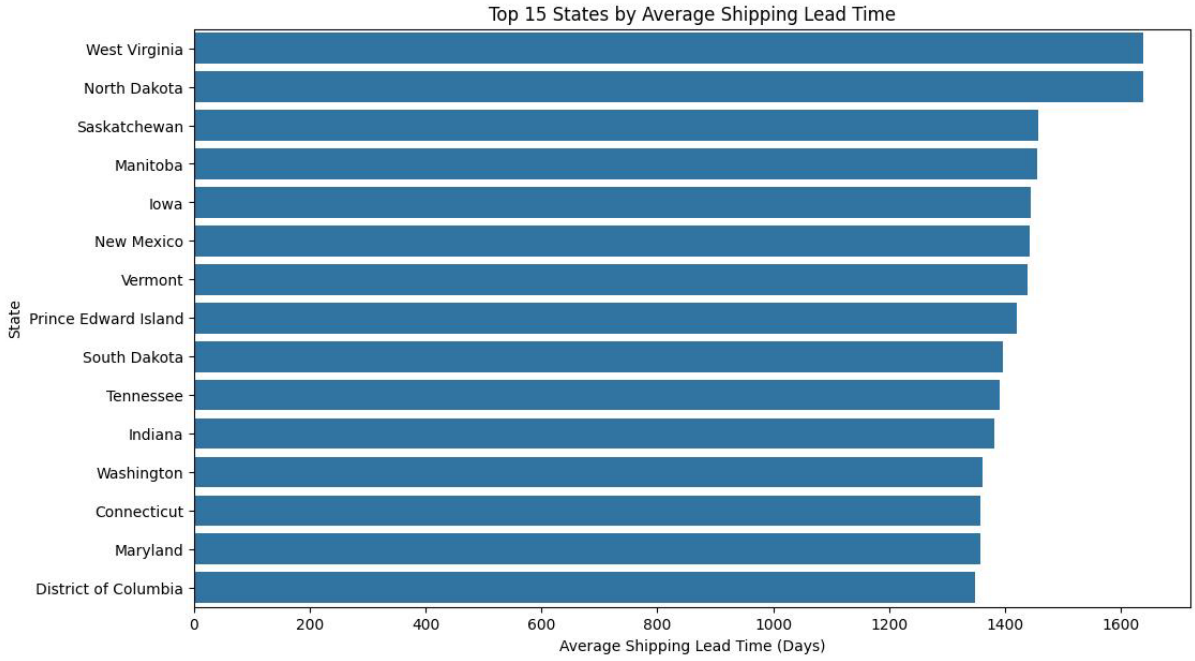




5.7 Geographic Analysis

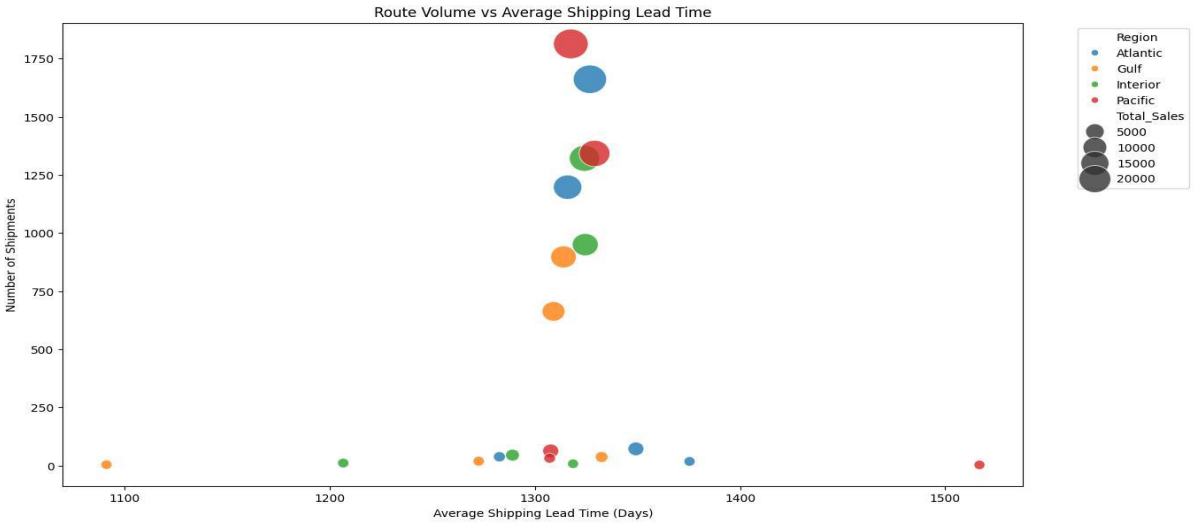
The notebook creates state/province-level summaries of shipment volume, orders, sales, gross profit and calculated lead time. Because the source field includes U.S. states and Canadian provinces, the geographic analysis is retained using the original State/Province terminology. The top geographic units by sales are shown in the notebook figure set.


Top 15 States by Sales
California
New York
Texas
Pennsylvania
Washington
Illinois
Ohio
Florida
Arizona
North Carolina
Michigan
Virginia
Georgia
Colorado
Tennessee
State
0
5000
10000
15000
20000
25000
Total Sales


Top 15 States by Average Shipping Lead Time
West Virginia
North Dakota
Saskatchewan
Manitoba
Iowa
New Mexico
Vermont
Prince Edward Island
South Dakota
Tennessee
Indiana
Washington
Connecticut
Maryland
District of Columbia
State
0
200
400
600
800
1000
1200
1400
1600
Average Shipping Lead Time (Days)

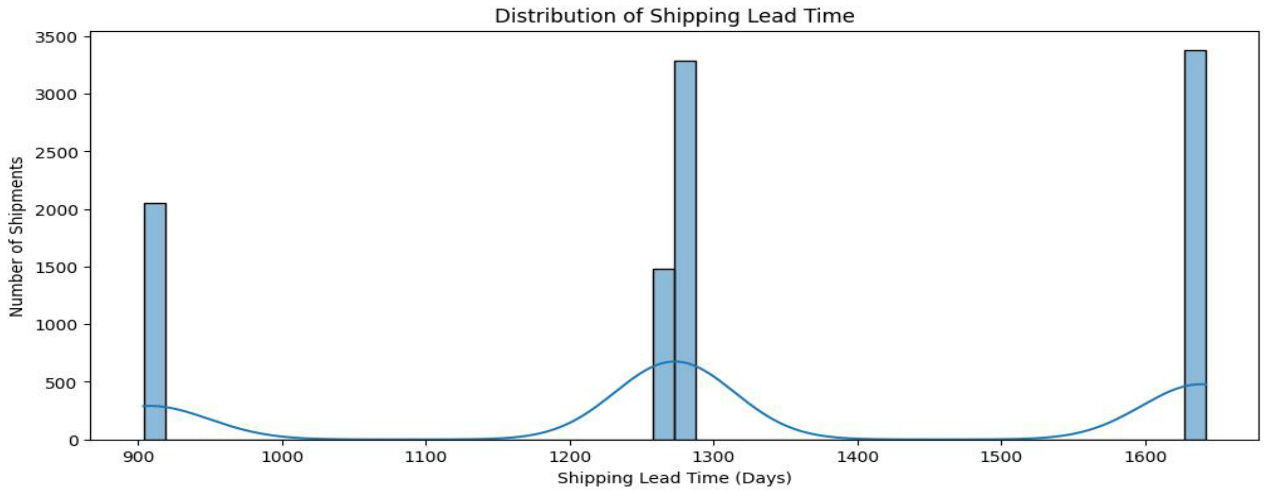
5.8 Route Analysis

Factory × Region combinations provide a route-level analytical unit. The largest shipment volumes occur in Lot's O Nuts routes to Pacific and Atlantic, followed by other Lot's O Nuts and Wicked Choccy's combinations. The route table combines commercial measures with lead-time statistics and delay frequency.

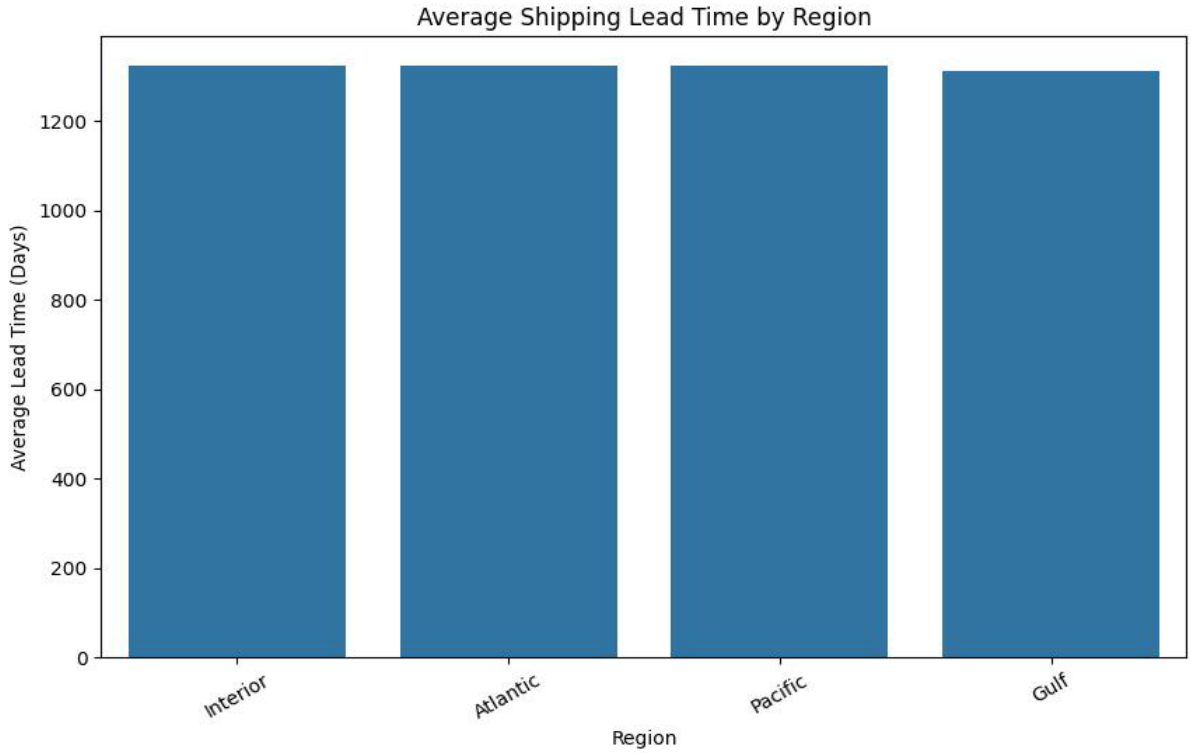


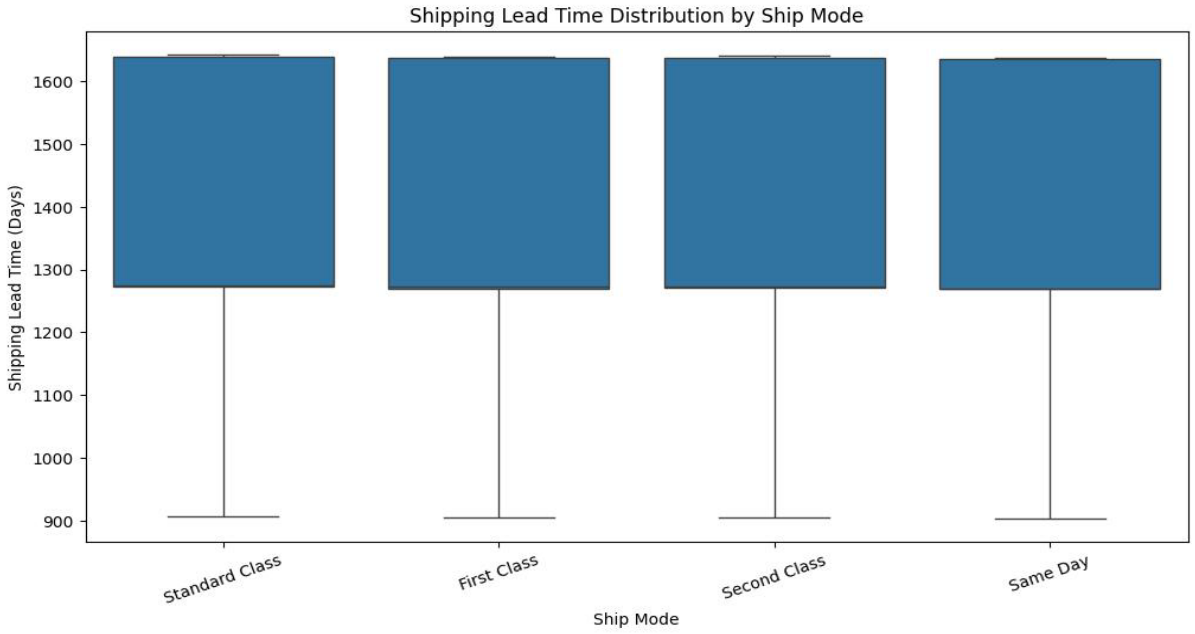
5.9 Shipping Lead-Time Data Quality

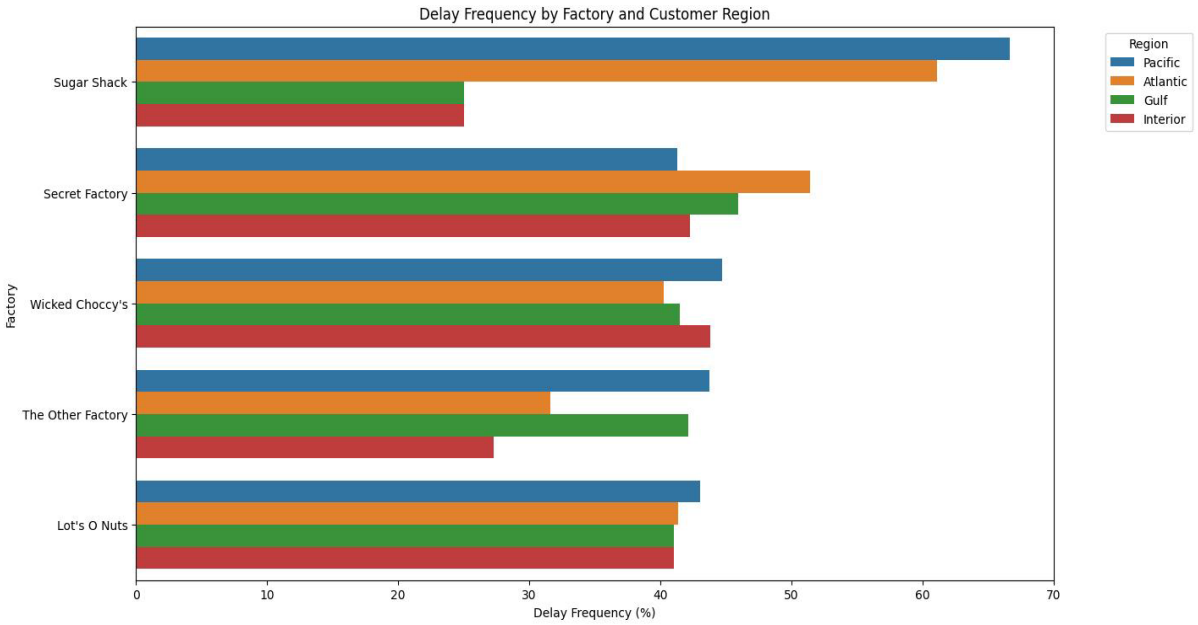
The most important analytical limitation is the Shipping Lead Time distribution. The minimum is 904 days, maximum 1642 days, mean 1320.84 days and median 1274 days. All 10,194 observations exceed 365 days. No negative values were identified. This is materially different from a conventional order-to-shipment KPI and therefore requires validation of the source date definitions before operational use.

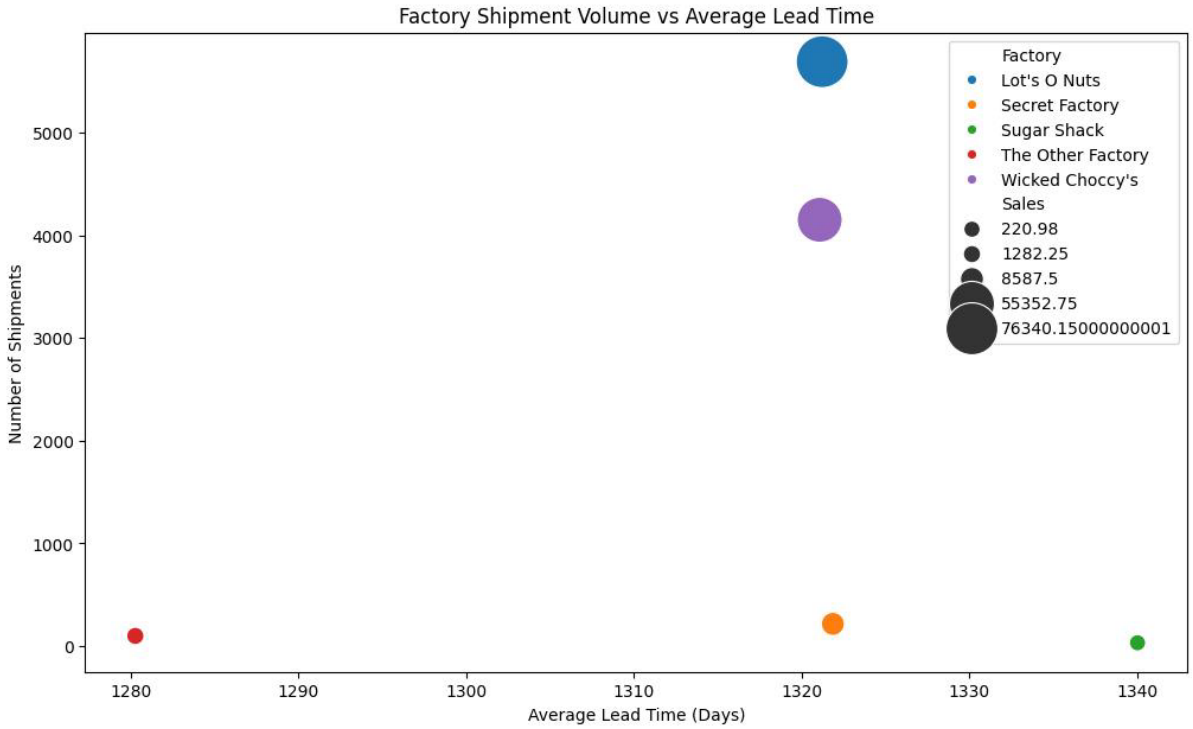



Average Shipping Lead Time by Factory
The Other Factory
Wicked Choccy's
Lot's O Nuts
Secret Factory
Sugar Shack
Factory
0
200
400
600
800
1000
1200
1400
Average Shipping Lead Time (Days)


Average Shipping Lead Time by Region
Average Lead Time (Days)
0
200
400
600
800
1000
1200
Interior
Atlantic
Pacific
Gulf
Region


Shipping Lead Time Distribution by Ship Mode
Shipping Lead Time (Days)
1600
1500
1400
1300
1200
1100
1000
900
Standard Class
First Class
Second Class
Same Day
Ship Mode


Delay Frequency by Factory and Customer Region
Region
Pacific
Atlantic
Gulf
Interior
Factory
Sugar Shack
Secret Factory
Wicked Choccy's
The Other Factory
Lot's O Nuts
0
10
20
30
40
50
60
70
Delay Frequency (%)



6 Discussion

The results demonstrate that the supplied dataset supports a multi-level descriptive analytics framework. Commercial measures establish the overall baseline, while product, factory, region, shipping-mode and geographic views reveal how that baseline is distributed.

Route analysis adds an operational perspective by connecting the project-defined factory mapping to customer regions. Because the route unit is an analytical combination rather than a confirmed physical route, the results should be used to prioritize investigation rather than to assign operational responsibility.

The most important methodological observation is that data completeness and operational validity are different concepts. The dataset contains no missing values and no duplicate rows, but the date-derived lead-time measure is unusually large for every record. The analysis therefore avoids converting this pattern into a claim about actual delivery delays.

The dashboard supports this interpretation by bringing commercial KPIs and operational drill-down into one interface. Users can inspect the same metrics used in the notebook while retaining the data-quality warning as part of the interpretation.

7 Limitations

• The factory assignments are project-defined and are not original dataset fields.

• The calculated Shipping Lead Time is not a validated service-level measure because the date semantics require confirmation.

• The 365-day quality threshold and median-based delayed flag are exploratory rules, not contractual business definitions.

• The State/Province field contains multiple geographic types and should not be interpreted as U.S. only data.

• The study is descriptive and diagnostic and does not establish causality.

• The results are specific to the supplied dataset and should not be generalized without additional data validation and a defined sampling frame.

8 Conclusion

This research developed a complete data analytics workflow for Nassau Candy distribution data, including data preparation, feature engineering, exploratory analysis, route analysis and interactive visualization. The dataset contains 10,194 records and supports a commercial baseline of $141,783.63 in sales and $93,442.80 in gross profit.

The central finding is that the calculated Shipping Lead Time is systematically large, ranging from 904 to 1642 days. Rather than treating this as validated delivery performance, the project documents it as a data-quality issue requiring source-system confirmation. Subject to that limitation, the factory, regional, product, shipping-mode and route analyses provide a structured descriptive framework for further investigation.

9 Future Scope

• Validate Order Date and Ship Date definitions against the authoritative operational system.

• Validate the product-to-factory mapping and replace it with official master data if available.

• Define formal service-level targets and replace exploratory thresholds with business-approved measures.

• Add time-series forecasting once temporal fields are validated.

• Develop anomaly detection for unusual orders and factory-region combinations.

• Extend the dashboard with customer-level profitability and segmentation.

• Implement a controlled data-refresh and deployment process for the Streamlit application.

References

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Few, S. (2006). Information Dashboard Design: The Effective Visual Communication of Data. O'Reilly Media.

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Tukey, J. W. (1977). Exploratory Data Analysis. Addison-Wesley.

Nassau Candy Distributor dataset. Supplied project dataset used for the internship analysis.

Nassau Candy Analysis notebook. Supplied project notebook containing preprocessing, exploratory analysis, route analysis and visualizations.